ENGINEERING PORTFOLIO

FALL 2026

# ETHAN WATKINS.

## Engineering & Physics · Centre College · Class of 2029

Structural and mechanical design, from a client's problem to a CAD model to a printed, tested part. Seeking a summer 2027 engineering internship.

INSIDE


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03 Endgame Climbing Mechanism
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06 DJ Copilot · Scoreboard
07 Research & Writing
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# 02

CLIENT DESIGN PROJECT

# Mini-Ecosystem Storage System

CLIENT
Prof. O'Quin, Centre College Biology Dept.

ROLE
CAD designer, EGR 110 team project

TOOLS
Fusion 360 · 3D printing

STATUS
Fall 2026 · final product due Nov.

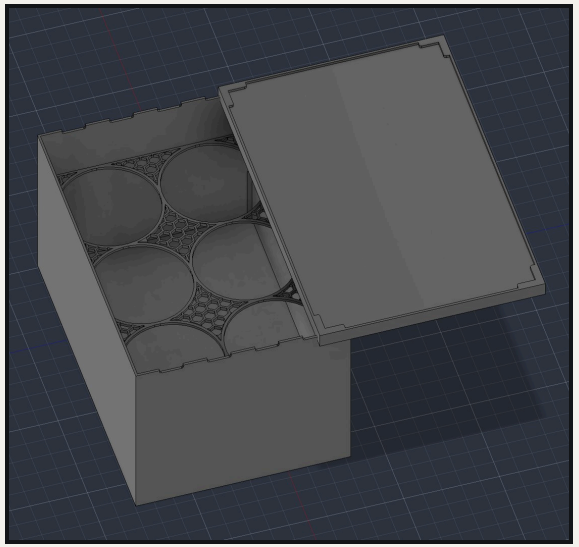
## Stackable storage for 30 living jar ecosystems, designed to be cheap to print.

### THE PROBLEM

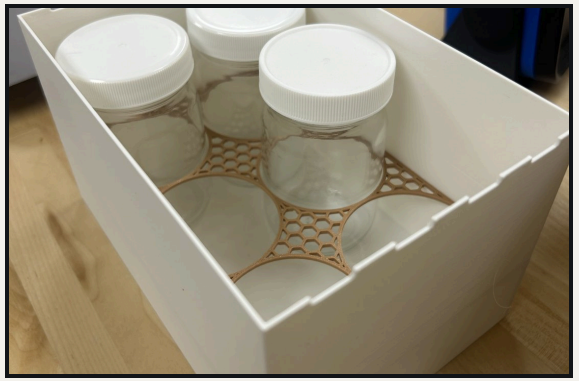
The biology department needed a way to hold and stack its mini-ecosystem jars with airflow. It also had to be food safe, cost-effective and easy to clean.

### WHAT I DESIGNED

- 5 modular boxes holding 6 jars each, modeled in Fusion 360
- A honeycomb rack with circular collars to seat each jar. Honeycomb uses less material, which keeps print cost low.
- A vented lid to keep air moving
- Interlocking features so stacked boxes lock together



Fusion 360 render, lid open



The printed box and insert

DESIGN PROCESS

| 01 | 02 | 03 | 04 | 05 |
|---|---|---|---|---|
| Client meeting | Decision matrix | Concept sketches | Prototype + client review | 3D-printed product |

# 03

COMPETITION ROBOTICS

# Endgame Climbing Mechanism

TEAM
VEX Robotics Team 4104A

ROLE
Team lead engineer and driver; designed the climbing system

TOOLS
Autodesk Inventor · VEX V5 · pneumatics

RESULT
2025 VEX Robotics World Championship



## A climb that lifted the whole robot, reliably, all the way to the World Championship.

### THE PROBLEM

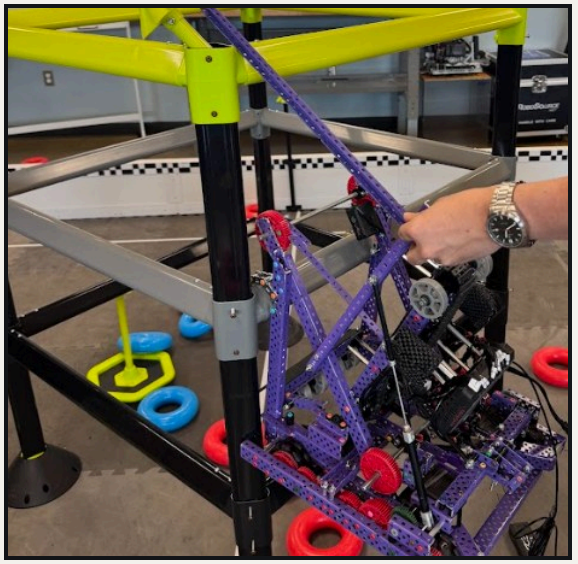
Lift the whole robot up the tower within VEX's 8-motor limit. A separate climber didn't have the torque.

### WHAT I DID

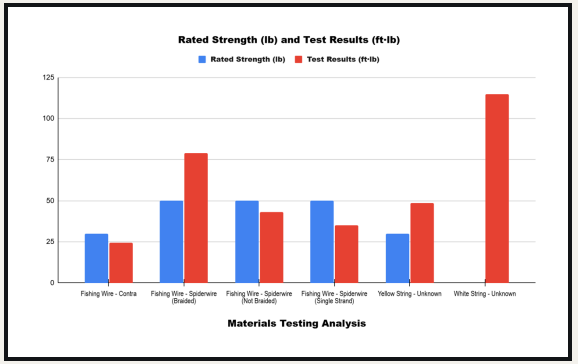
- Led the team as lead engineer and drove in matches
- Built the climbing winch into the drivetrain
- Designed a pneumatic-piston PTO clutch that shifts the drive motors into low gear for maximum torque
- Reinforced the lift and reshaped the hooks
- Chose the winch line from force-gauge tests

### RESULT

Consistent climbs in elimination matches at the 2025 World Championship.



Testing the climber on the tower




Force-gauge tests of six climbing lines

### ITERATION

| V1 | PTO | GEAR | FINAL |
|---|---|---|---|
| Separate climber lacked torque | Pneumatic clutch shares drive motors | Low gear for max torque | Load-tested, climbing at Worlds |

# 04

PRODUCT · SIDE BUSINESS

# NFC Phone-Lock Tag

ROLE
Designer, maker and seller

WHEN
2026

TOOLS
3D printing · NFC · iOS Shortcuts · Foqos

SALES
8 sold at $10 · about 12¢ each to make

## A physical off-switch for distracting apps. Tap to lock, tap again to unlock.

### THE PROBLEM

Software app blockers are easy to switch off. A physical tag you have to walk to works better, so I made a lower-cost version I could build and sell.

### WHAT I BUILT

- A 3D-printed tag housing with an embedded NFC chip
- The lock flow on Foqos, a free app that blocks apps through Apple's Screen Time
- A one-tap Shortcuts automation, so a single tap locks or unlocks without opening an app
- A 3-page customer setup guide with a troubleshooting table

### RESULT

Sold 8 tags at $10 each, on a production cost of about 12¢ per tag.




A finished tag



Setup Guide: Your Phone Lock Tag

Oct 3, 2026 · @Ethan

Tap your phone to the tag to lock distracting apps, and tap again to unlock them. Setup takes about 5 minutes.

What you need

- **An iPhone with NFC** (iPhone XS or newer). This setup does not work on Android.
- **Your tag.** It comes ready to use; nothing needs to be written to it.
- **The free Foqos app** from the App Store. It does the app blocking using Apple's built-in Screen Time system.

Setup

1. **Install Foqos.** Search "Foqos" in the App Store and download it (free).
2. **Allow Screen Time access.** Open Foqos and tap **Allow** when it asks for Screen Time permission. Blocking won't work without it.
3. **Create a profile.** Tap to add a new profile and give it a name, like "Focus" or "Phone Down."
4. **Choose apps to block.** Pick the apps and websites you want locked, such as social media, games, or video apps.
5. **Set the blocking strategy to NFC tag.**
6. **Scan your tag.** Under **Physical Unlocks**, tap to add a tag, then hold the top edge of your iPhone against the tag until it confirms.
7. **Turn on Prevent App Deletion.** This stops you from deleting Foqos to get around the

The customer setup guide I wrote

### HOW IT WORKS

| TAP | WALK AWAY | TAP AGAIN |
|---|---|---|
| Chosen apps lock | Tag stays in another room | Only the tag unlocks |

# 05

SOFTWARE · STUDY TOOL

# Clicky: AI Physics Whiteboard

BUILT FOR
Studying for my Electricity & Magnetism class

ROLE
Creator

WHEN
Fall 2026

TECH
Windows · Claude API · Kokoro + Piper neural voices

## A study tutor that draws on my homework and talks me through it.

### WHAT IT DOES

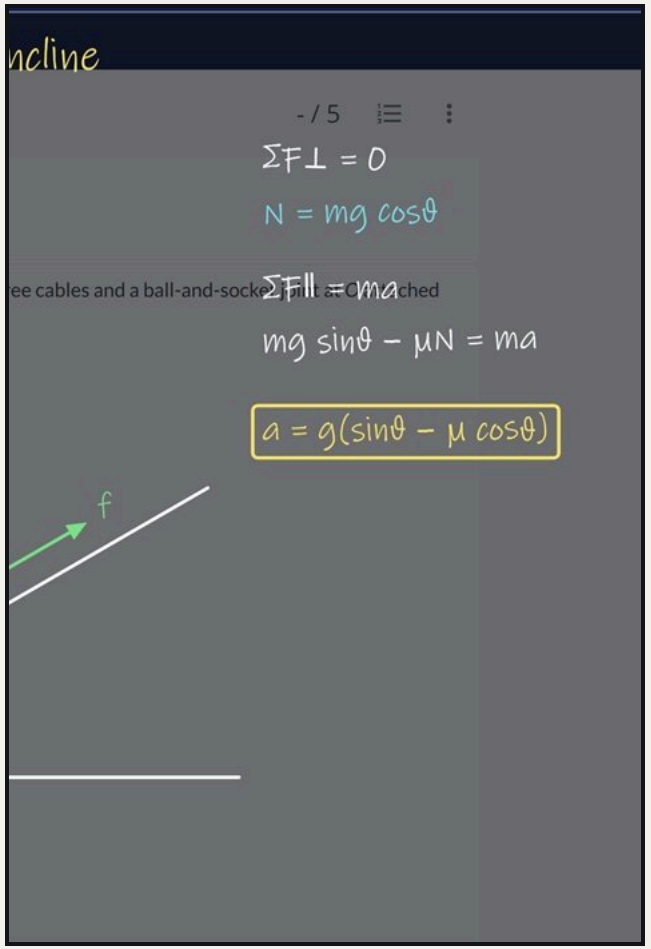
Point it at a problem on screen. It draws free-body diagrams and equations over the page in chalk-style ink while a voice explains each step.

### HOW I BUILT IT

- Each lesson step pairs a spoken line with drawing commands: arrows, lines, circles, equations
- The voice starts speaking while the lesson is still being written
- Speech runs offline on the laptop
- Step slider, pause and replay

### RESULT

I use it for E&M and Statics, such as solving a 13-member truss joint by joint.




Clicky working a block on a rough incline: forces, then the acceleration

PIPELINE

| 01 CAPTURE | 02 PLAN | 03 SPEAK | 04 DRAW |
|---|---|---|---|
| Lock onto the homework window | Claude writes the lesson | Local voice reads each step | Chalk overlay on screen |

# 06

SOFTWARE · AI DJ ASSISTANT

# DJ Copilot

BUILT FOR

DJing chapter events as fraternity Social Chair

WHEN

Fall 2026 · creator

TECH

PowerShell · C# · JavaScript · SQLite · Claude API · Windows OCR

ALSO HERE

Scoreboard controller · Arduino + C++

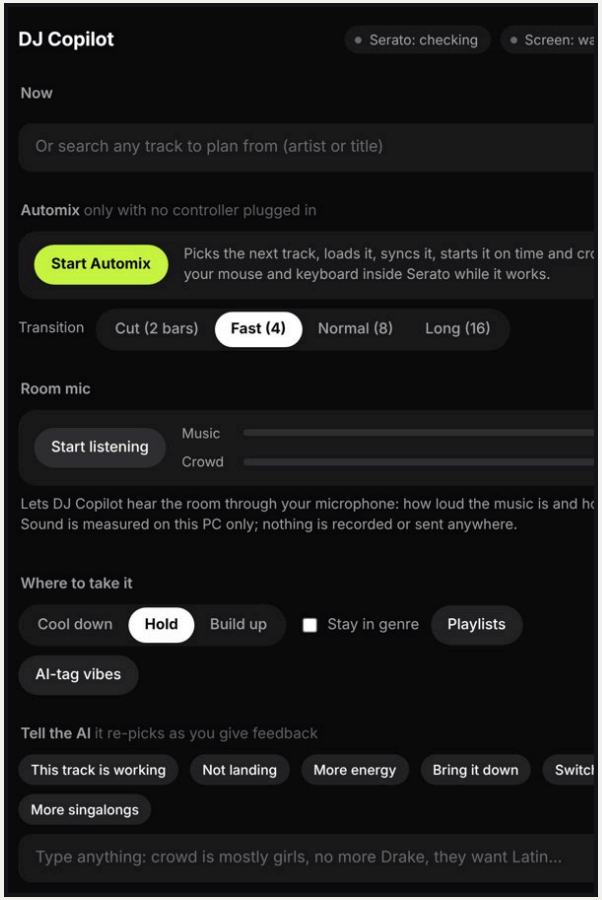
## An AI co-DJ that picks the next track and can mix it in by itself.

### HOW IT WORKS

- Reads Serato's library and play history, keeps tracks that match in tempo and key, and has Claude rank the best five next
- Crowd buttons like "Not landing" or "More energy" re-steer the picks
- Automix reads Serato's screen, loads and syncs the next track, then crossfades

### RESULT

Two automatic transitions back to back, tested in Serato's practice mode.




DJ Copilot's control panel

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### EMBEDDED · C++

## Scoreboard Controller

Arduino firmware drives a four-digit, seven-segment scoreboard through MOSFETs. A Windows C++ app sends scores over serial.

```
> 12 8
Sent: SET 12 8
Arduino: OK   HOME 12 · AWAY 08
```

# 07

TECHNICAL COMMUNICATION

## Research & Writing

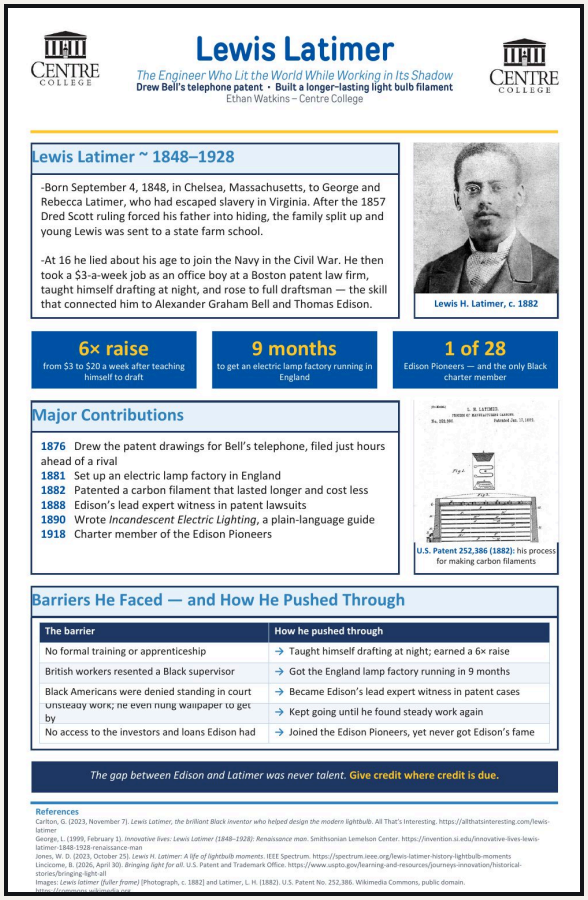
COURSES

EGR 110 Engineering · CHE 131 Chemistry

SKILLS

Cited research · poster design · lab analysis

# Explaining engineering clearly, on paper and on a poster.



Lewis Latimer

*The Engineer Who Lit the World While Working in Its Shadow*

**Drew Bell's telephone patent · Built a longer-lasting light bulb filament**

Ethan Watkins – Centre College

**Lewis Latimer ~ 1848–1928**

-Born September 4, 1848, in Chelsea, Massachusetts, to George and Rebecca Latimer, who had escaped slavery in Virginia. After the 1857 Dred Scott ruling forced his father into hiding, the family split up and young Lewis was sent to a state farm school.

-At 16 he lied about his age to join the Navy in the Civil War. He then took a $3-a-week job as an office boy at a Boston patent law firm, taught himself drafting at night, and rose to full draftsman — the skill that connected him to Alexander Graham Bell and Thomas Edison.

Lewis H. Latimer, c. 1882

**6× raise** — from $3 to $20 a week after teaching himself to draft

**9 months** — to get an electric lamp factory running in England

**1 of 28** — Edison Pioneers — and the only Black charter member

**Major Contributions**

1876 Drew the patent drawings for Bell's telephone, filed just hours ahead of a rival
1881 Set up an electric lamp factory in England
1882 Patented a carbon filament that lasted longer and cost less
1888 Edison's lead expert witness in patent lawsuits
1890 Wrote *Incandescent Electric Lighting*, a plain-language guide
1918 Charter member of the Edison Pioneers

**U.S. Patent 252,386 (1882):** his process for making carbon filaments

**Barriers He Faced — and How He Pushed Through**

| The barrier | How he pushed through |
|---|---|
| No formal training or apprenticeship | → Taught himself drafting at night; earned a 6× raise |
| British workers resented a Black supervisor | → Got the England lamp factory running in 9 months |
| Black Americans were denied standing in court | → Became Edison's lead expert witness in patent cases |
| Unsteady work; he even hung wallpaper to get by | → Kept going until he found steady work again |
| No access to the investors and loans Edison had | → Joined the Edison Pioneers, yet never got Edison's fame |

*The gap between Edison and Latimer was never talent.* **Give credit where credit is due.**

References


Carlton, G. (2023, November 7). *Lewis Latimer, the brilliant Black inventor who helped design the modern lightbulb.* All That's Interesting. https://allthatsinteresting.com/lewis-latimer

George, L. (1999, February 1). *Innovative lives: Lewis Latimer (1848–1928): Renaissance man.* Smithsonian Lemelson Center. https://invention.si.edu/innovative-lives-lewis-latimer-1848-1928-renaissance-man

Jones, W. D. (2023, October 25). *Lewis H. Latimer: A life of lightbulb moments.* IEEE Spectrum. https://spectrum.ieee.org/lewis-latimer-history-lightbulb-moments

Lincicome, B. (2026, April 30). *Bringing light for all.* U.S. Patent and Trademark Office. https://www.uspto.gov/learning-and-resources/journeys-innovation/historical-stories/bringing-light-all

Images: *Lewis latimer (fuller frame)* (Photograph, c. 1882) and Latimer, L. H. (1882). U.S. Patent No. 252,386. Wikimedia Commons, public domain. https://commons.wikimedia.org


My 11×17 research poster

EGR 110 · PAPER + POSTER

## Lewis Latimer: The Engineer Who Lit the World While Working in Its Shadow

I researched the self-taught draftsman who drew the patent drawings for Bell's telephone and patented a longer-lasting carbon filament. I wrote the paper and designed the poster from cited sources, including IEEE Spectrum, the Smithsonian and the USPTO.

CHE 131 · LAB REPORT

## Iron in a Multivitamin, by Spectrophotometry

I measured a tablet's iron by reading the Fe–phenanthroline complex at 510 nm against a Beer–Lambert calibration curve. I found about 11 mg per tablet against an 18 mg label, a 39% error.

08

SKILLS + CONTACT

# Toolkit & Contact

## At home in CAD, in the shop and on a job site.

### CAD

Autodesk Inventor (Certified User) · Fusion 360 · Onshape · Revit

### FABRICATION

3D printing · prototyping · mechanical assembly · welding · soldering

### PROGRAMMING

C · C++ · Arduino · Microsoft Excel

### CERTIFICATIONS

Autodesk Certified User: Inventor (2023) · EOP Engineering Exam (2024)

### COURSEWORK

Statics · Electricity & Magnetism · Physics: Mechanics · Calculus II & III · Engineering I & II · Civil Engineering & Architecture

### BUILDING IN THE REAL WORLD

2019–2023 · FAMILY CONTRACTING

Built decks and gazebos from structural layout to completion

HABITAT FOR HUMANITY

Helped plan blueprints and build budget-friendly housing


LET'S TALK

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